

CS 101: Fundamentals of Computer Science I

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Introduction

- About me
 - Ph.D. in Computer Science, *Virginia Tech*
 - Research: AI-based Smart Systems, Machine Learning, Explainable AI
- **About you**
 - Your name
 - Your major
 - Have you programmed before? If yes, which language(s)?
 - What do you think programming is?

About This Course

- What will you learn?
 - Think like a programmer
 - Solve problems systematically
 - Write C programs
 - Understand how computers execute programs
 - Build a foundation for later CS courses

This course is **NOT** about memorizing syntax.
It is about learning problem solving.

What is Programming?

- Programming is the process of writing instructions that tell a computer how to solve a problem.
- **Algorithm:** A step-by-step procedure for solving a problem.
- **Program:** An algorithm written in a programming language that a computer can execute.

Example: Odd or Even?

Problem

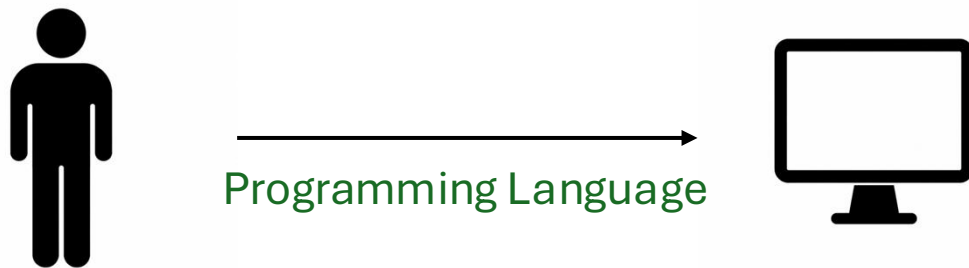
- Determine whether an integer is odd or even.

Algorithm

1. Input an integer **n**.
2. Divide **n** by **2**.
3. Check the remainder.
4. If the remainder is **0**, output "**Even**".
5. Otherwise, output "**Odd**".

What Is a Programming Language?

- How do we communicate with a computer?
 - Computers understand only machine instructions.
 - Humans write programs using programming languages.
 - A **programming language** lets us express an algorithm in a form that can later be translated into machine code.



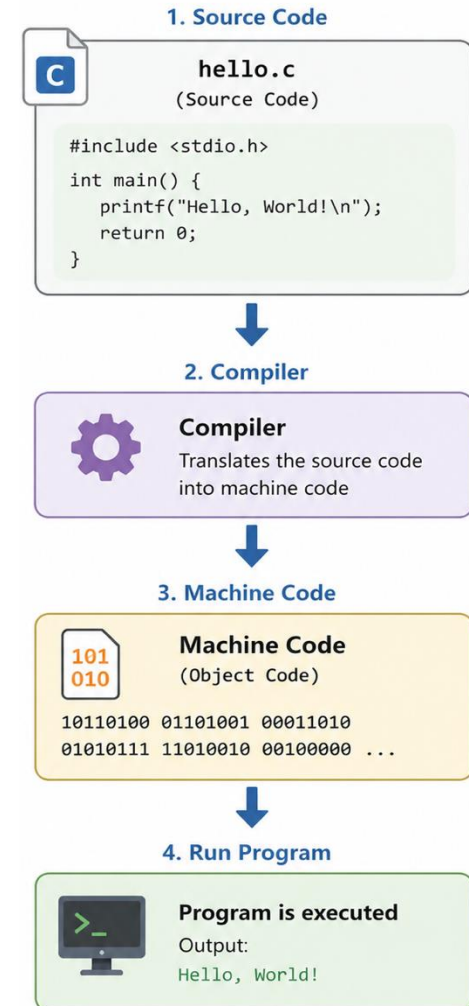
An algorithm becomes a program when it is written in a programming language.

Why Learn C?

- Why is C still important?
 - Foundation of many modern languages
 - Fast and efficient
 - Close to computer hardware
 - Used in
 - Operating systems
 - Embedded systems
 - Robotics
 - Microcontrollers
 - Device drivers

How Does the Computer Understand C?

- Computers understand only machine code (0s and 1s).
- A **compiler** translates your **C program** into **machine code** that the computer can execute.



Key Takeaways

- What programming is
- Algorithm vs. program
- Programming language
- Why C
- Compiler
- Executable